Supplementary Material for
Secretary-Style Split Search for Decision Trees

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Abstract

This supplementary document provides additional empirical figures supporting the results reported in the main manuscript. It includes Supplementary Figure S1 on size-stratified Pareto comparisons, Supplementary Figure S2 on the screening analysis of the parametric secretary strategy, Supplementary Figures S3–S5 on ridgeline summaries of within-dataset variability, Supplementary Figures S6–S7 and S12 on reliability and joint-success plots, Supplementary Figure S8 on predicted regime maps, Supplementary Figures S9–S11 on within-dataset whisker summaries, Supplementary Figure S13 on exhaustive gain-landscape diagnostics for the regression–classification asymmetry, and Supplementary Figures S14–S15 on split-search effort comparisons.

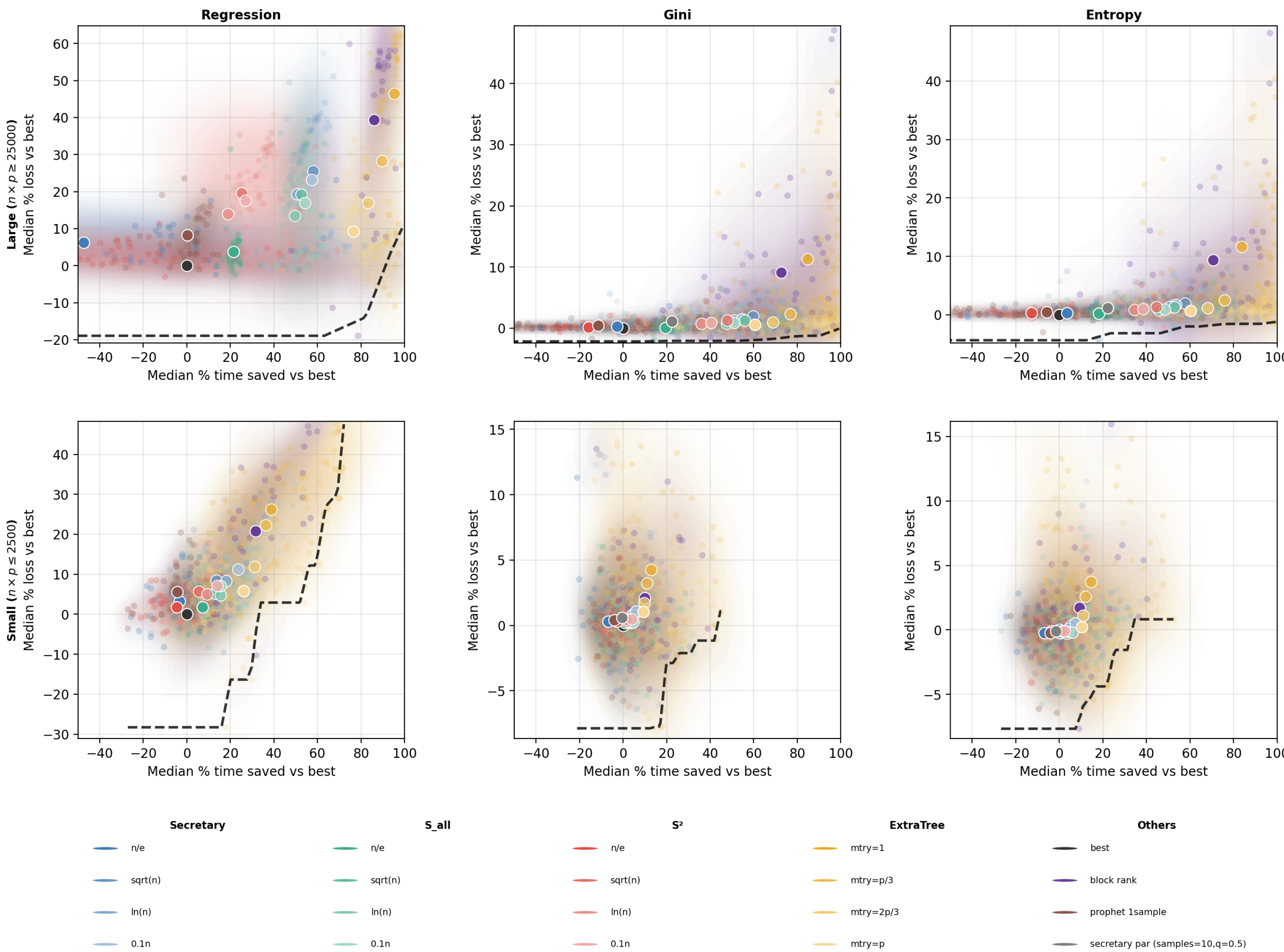

Figure S1: Size-stratified Pareto comparisons. The top row reports large datasets, defined by $n \times p \geq 25000$, and the bottom row reports small datasets, defined by $n \times p \leq 2500$. Columns correspond to regression, classification with Gini impurity, and classification with entropy impurity. In each panel, each faint point represents one benchmark dataset summarized by the median over 50 runs for a given method variant. The horizontal axis is the median percentage of training time saved relative to the exhaustive `scikit-learn` splitter, and the vertical axis is the median percentage loss relative to the same baseline. The black point at the origin is the exhaustive splitter. Large white-edged markers indicate the cross-dataset medians of the corresponding methods. The figure shows that the time–accuracy trade-off is substantially less favorable in regression than in classification, and that the behavior of the methods differs between small and large datasets.

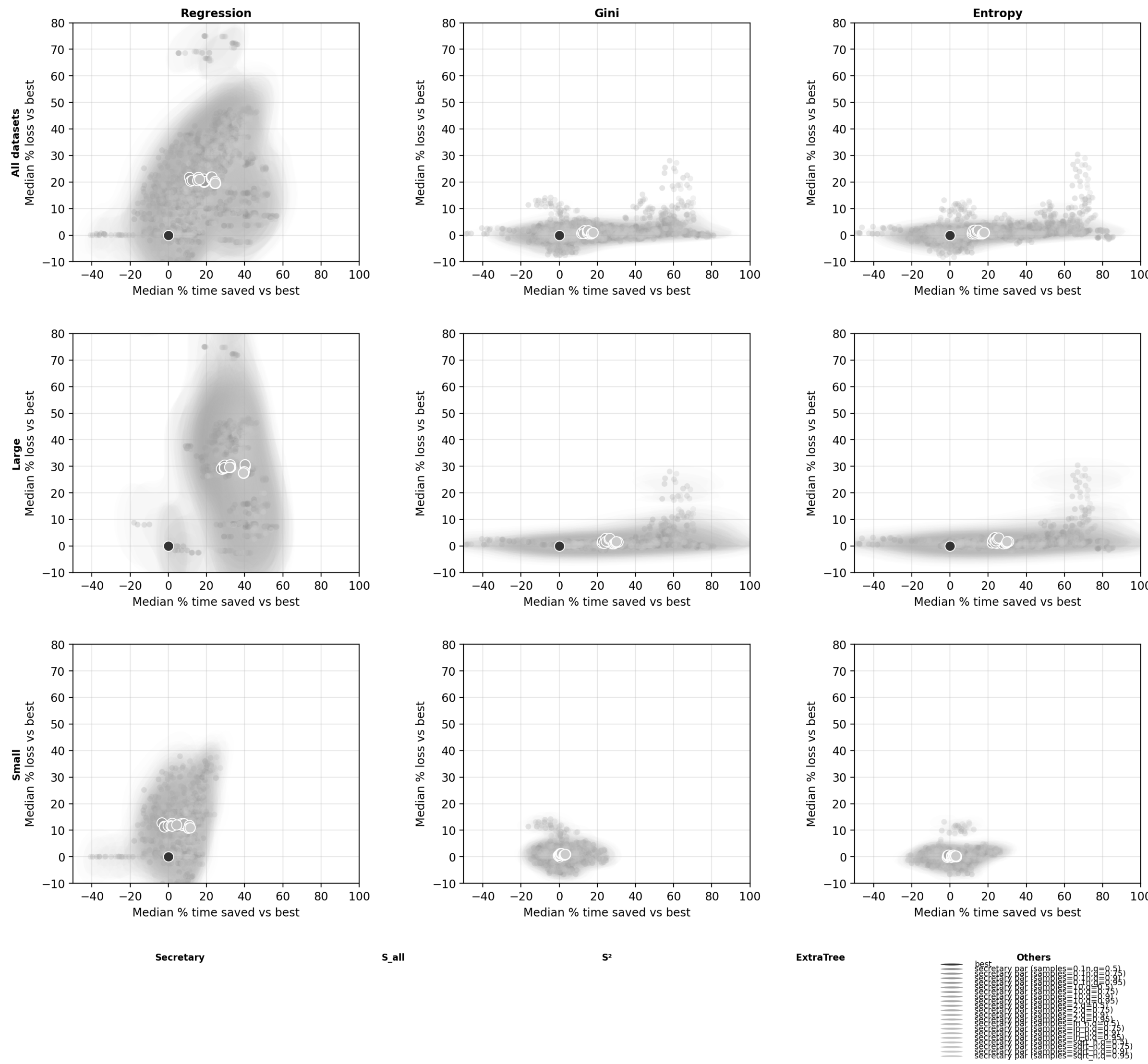

Figure S2: Screening analysis of the parametric secretary strategy S_{par}. Rows correspond to all datasets, large datasets, and small datasets. Columns correspond to regression, classification with Gini impurity, and classification with entropy impurity. Each colored point represents one tested configuration of S_{par}, defined by the number of gain samples used to fit the working distribution and by the acceptance quantile used for stopping. The horizontal axis reports the median percentage of training time saved relative to the exhaustive splitter, and the vertical axis reports the median percentage loss relative to the same baseline, both computed over benchmark datasets. The black point at the origin is the exhaustive baseline, and large white-edged markers indicate cross-dataset medians of the corresponding configurations. Most settings are globally dominated once fitting overhead is taken into account, but one low-overhead configuration remains competitive on small classification datasets; this is why a single representative S_{par} line is retained in the main classification figures and table.

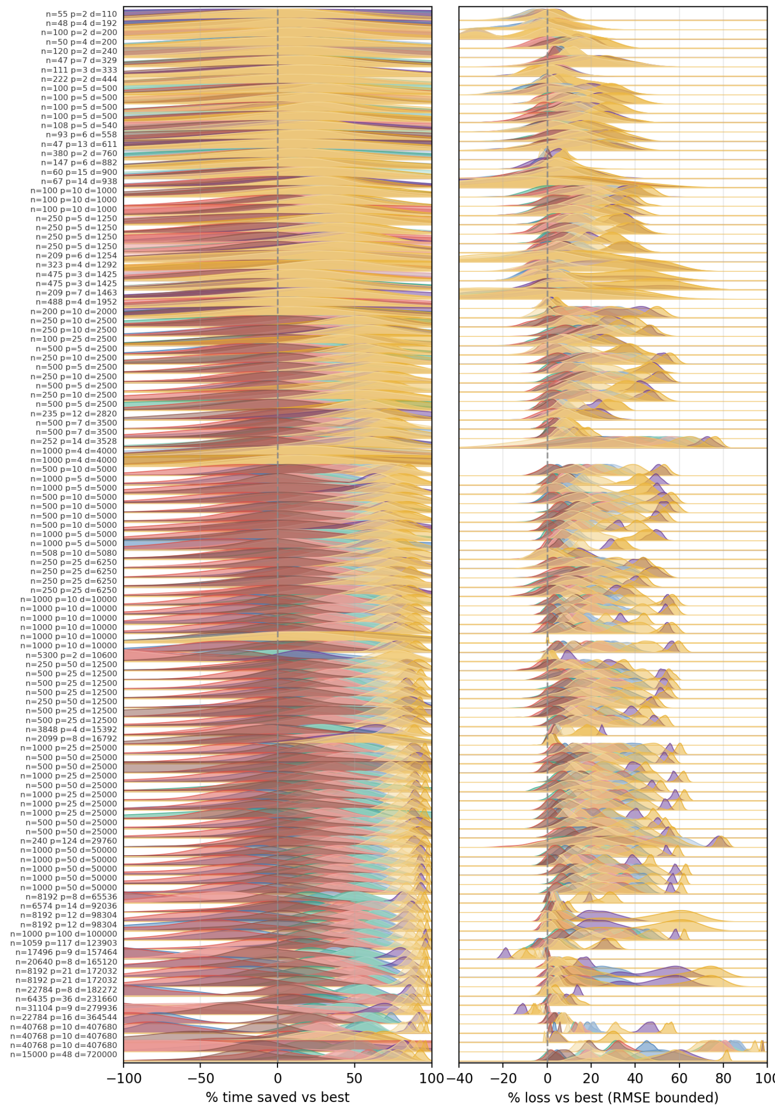

Figure S3: Ridgeline summaries for regression. Each horizontal ridge corresponds to one benchmark dataset, identified on the vertical axis by its sample size and number of features. The left panel shows the empirical distribution, over 50 runs, of the percentage of training time saved relative to the exhaustive splitter; the right panel shows the corresponding distribution of percentage RMSE loss relative to the same baseline. Colors identify methods. The dashed vertical line at zero marks parity with the exhaustive splitter. These ridgelines make explicit the within-dataset stochastic variability induced by randomized split exploration and show that the regression benchmark is dominated by large positive loss distributions for aggressive secretary-style rules.

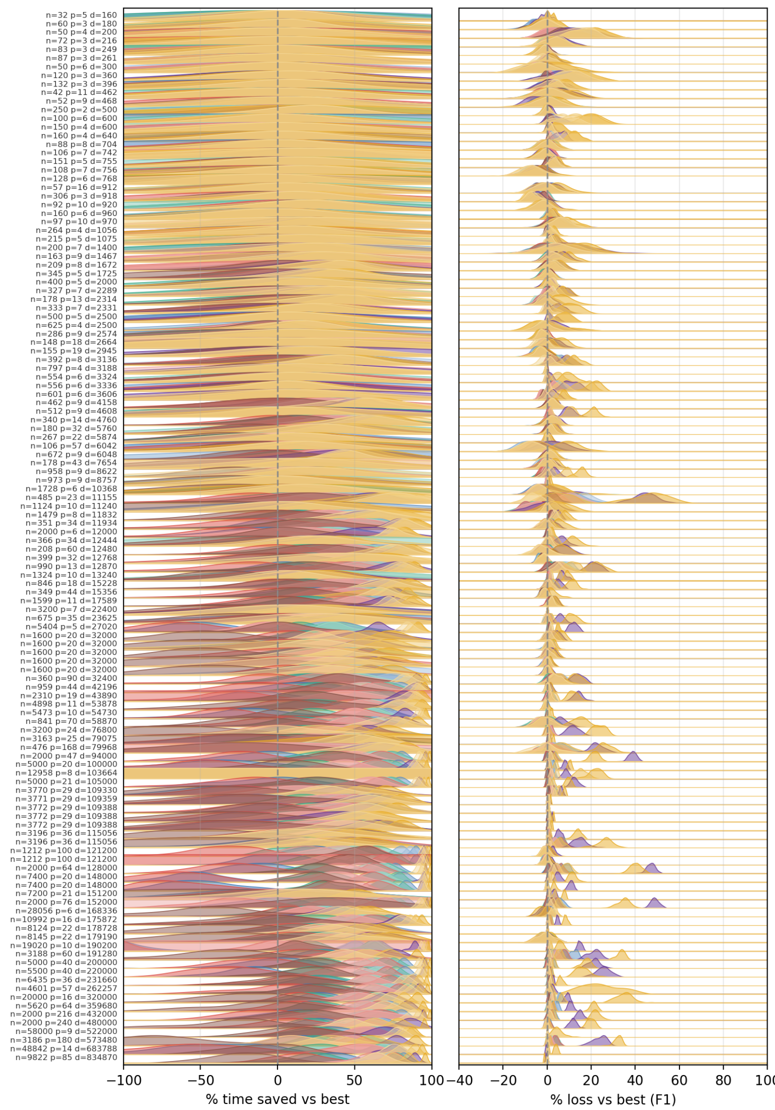

Figure S4: Ridgeline summaries for classification with Gini impurity. Each horizontal ridge corresponds to one benchmark dataset. The left panel shows the empirical distribution, over 50 runs, of the percentage of training time saved relative to the exhaustive splitter, and the right panel shows the corresponding distribution of percentage F1 loss relative to the same baseline. Colors identify methods, and the dashed vertical line at zero marks parity with the exhaustive splitter. Compared with regression, these ridgelines show a much larger concentration of runs near zero predictive loss, which helps explain why secretary-style stopping is more successful in classification.

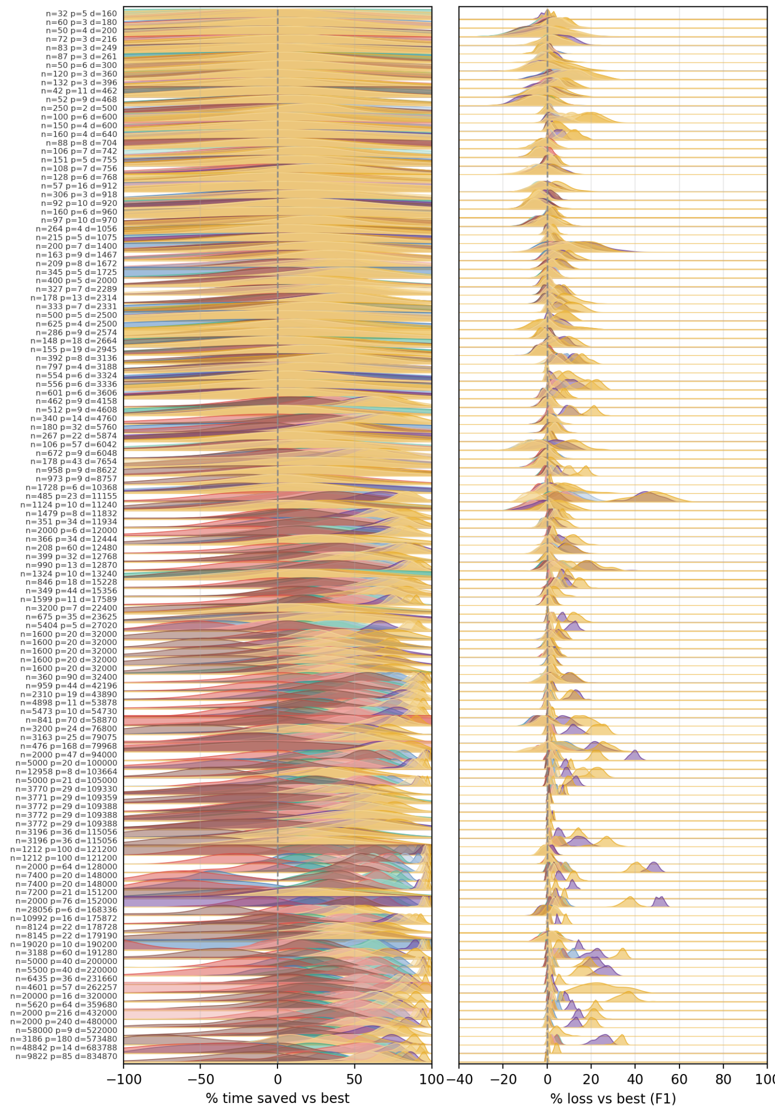

Figure S5: Ridgeline summaries for classification with entropy impurity. As in Supplementary Figure S4, each horizontal ridge corresponds to one benchmark dataset, the left panel shows percentage time saved, and the right panel shows percentage F1 loss, both relative to the exhaustive splitter and computed over 50 repeated runs. The figure confirms the broad similarity between Gini and entropy classification in the benchmark, while also revealing dataset-specific variability in the reliability of the different stopping rules.

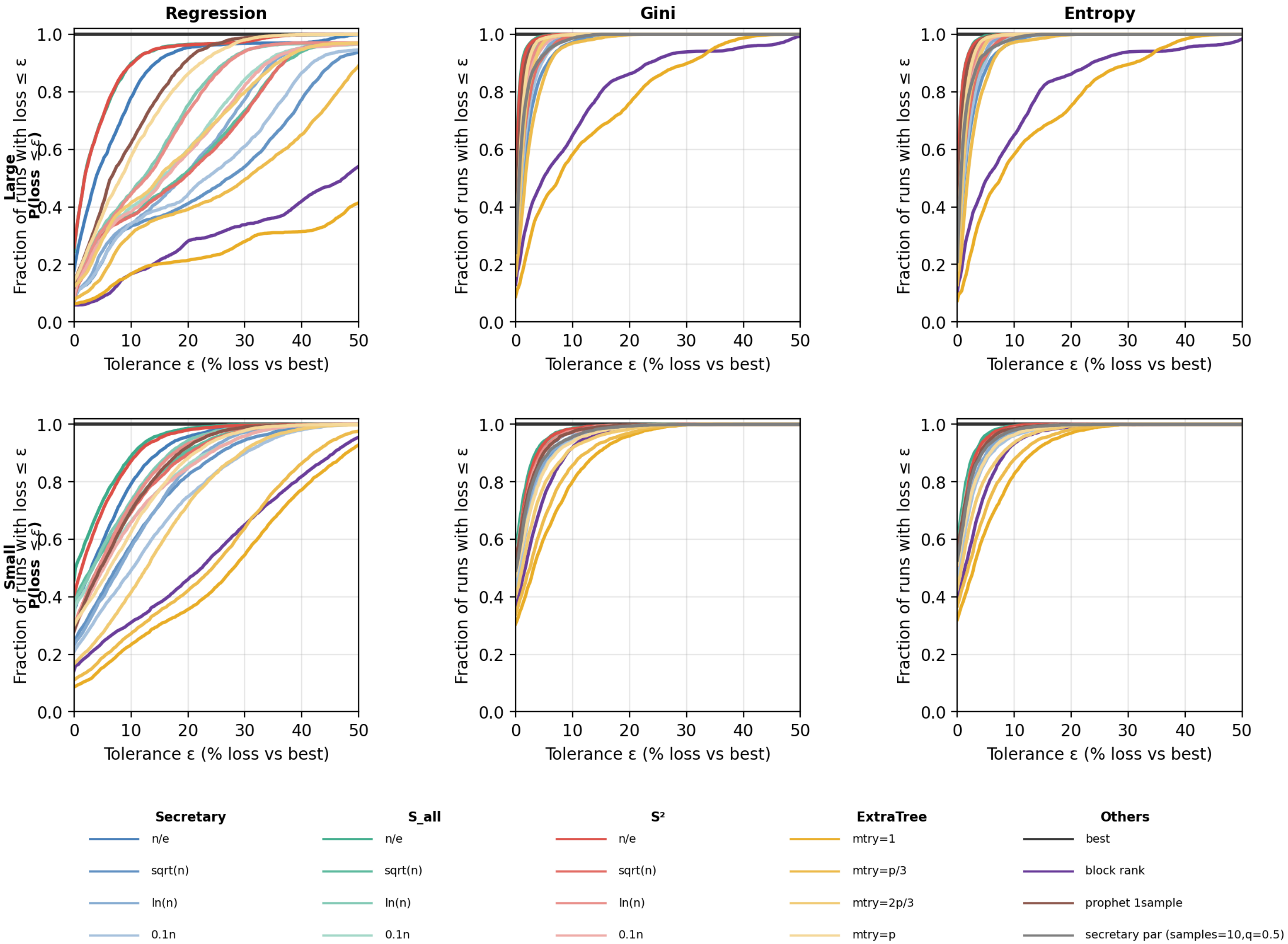

Figure S6: Loss-only success probabilities stratified by dataset size. The top row reports large datasets and the bottom row reports small datasets; columns correspond to regression, Gini classification, and entropy classification. In each panel, the horizontal axis is the predictive-loss tolerance ε, and the vertical axis is the empirical fraction of runs whose loss is at most ε, aggregated over the datasets in the corresponding size stratum. Colored curves identify methods, and the horizontal black line at one is the exhaustive baseline. These plots isolate predictive reliability independently of computational gains and show that classification remains much more tolerant to early stopping than regression in both size regimes.

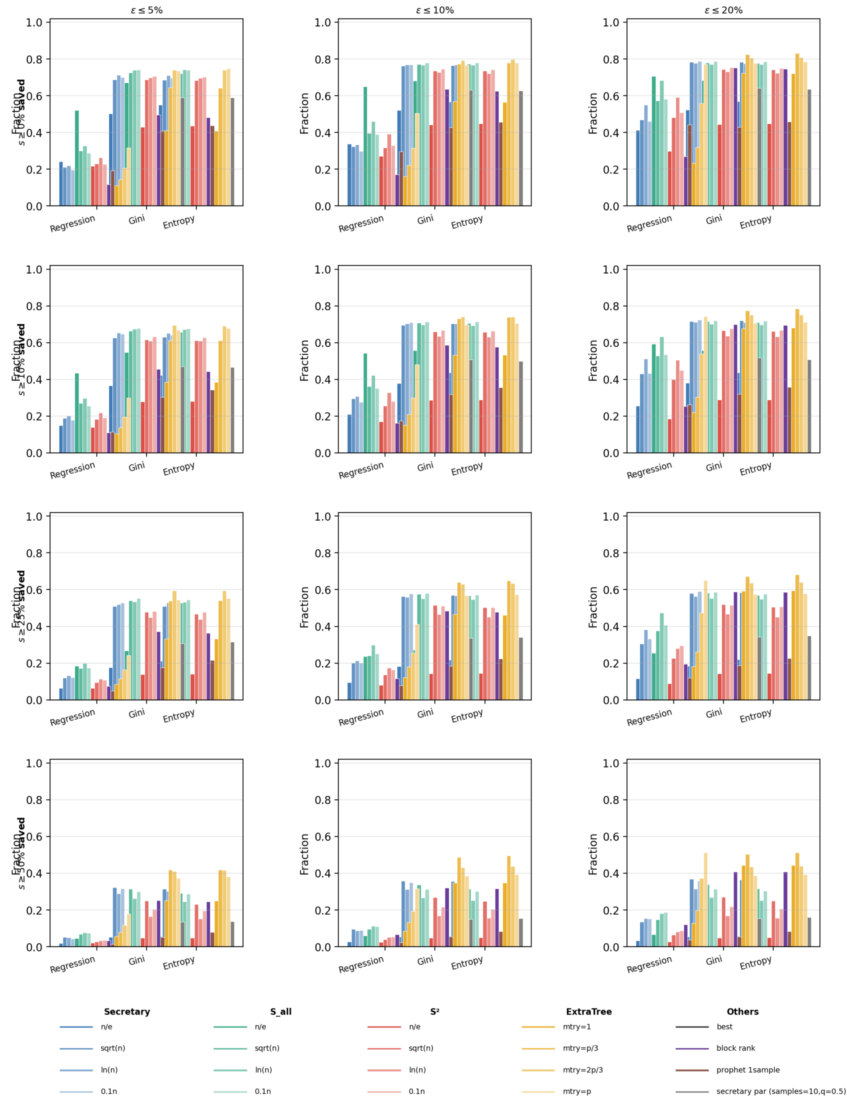

Figure S7: Joint success probabilities over all benchmark datasets. Columns correspond to predictive-loss tolerances $\varepsilon \in \{5\%, 10\%, 20\%\}$, and rows correspond to minimum time-saving thresholds $s \in \{0\%, 10\%, 25\%, 50\%\}$. In each panel, bar heights report the empirical fraction of runs satisfying the joint event "time saved $\geq s$ and predictive loss $\leq \varepsilon$", aggregated over all datasets for regression, classification with Gini impurity, and classification with entropy impurity. These summaries provide the full operating-point view underlying the compact success-probability discussion in the main text and show how the ranking of the methods changes as the speedup requirement and the admissible loss are varied.

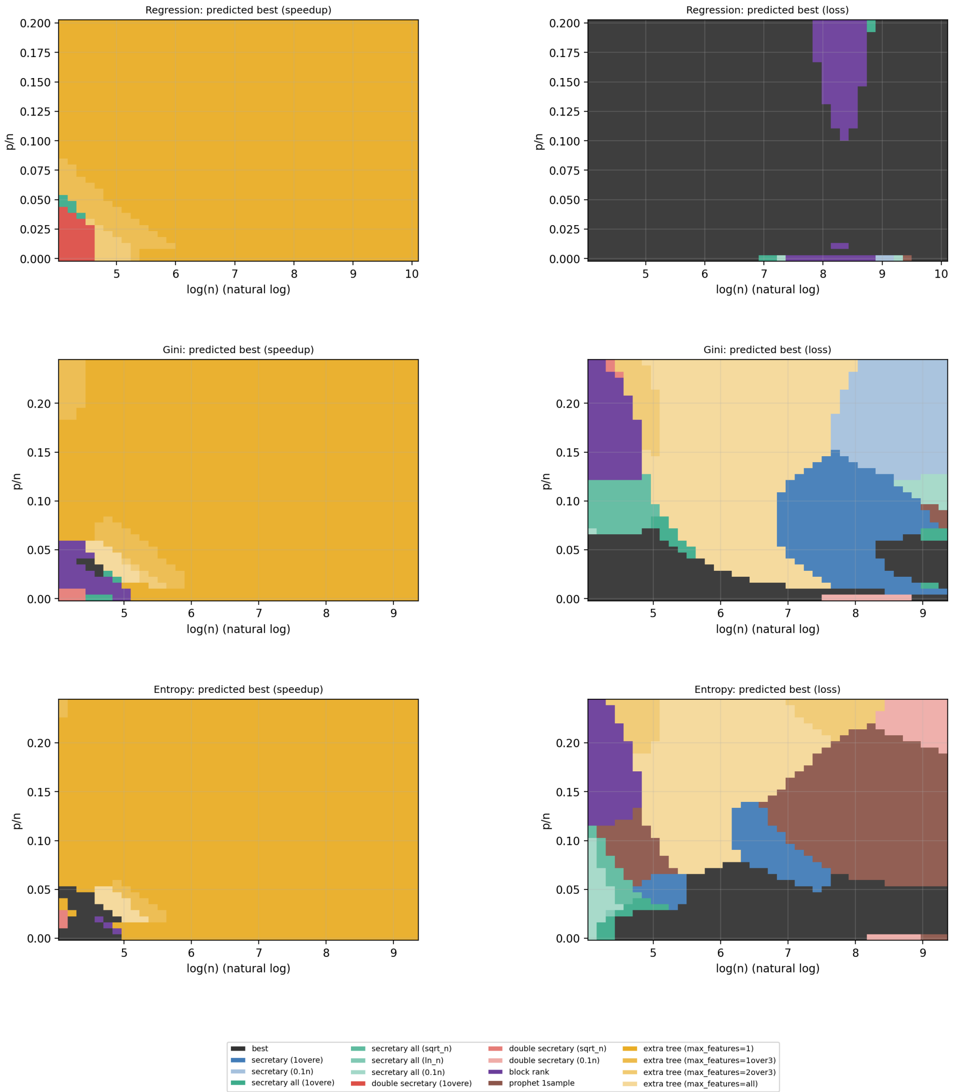

Figure S8: Predicted regime maps in the $(\log n, p/n)$ plane. Rows correspond to regression, Gini classification, and entropy classification. The left column shows the empirically predicted best method when the objective is to maximize speedup; the right column shows the empirically predicted best method when the objective is to minimize predictive loss. Background colors identify the method predicted to be best in each local region of the covariate plane. These figures summarize how the preferred split-search strategy changes with sample size and effective dimensionality. In regression, the exhaustive splitter dominates much of the loss-oriented map, whereas in classification the secretary-style and rank-adaptive rules occupy larger regions.

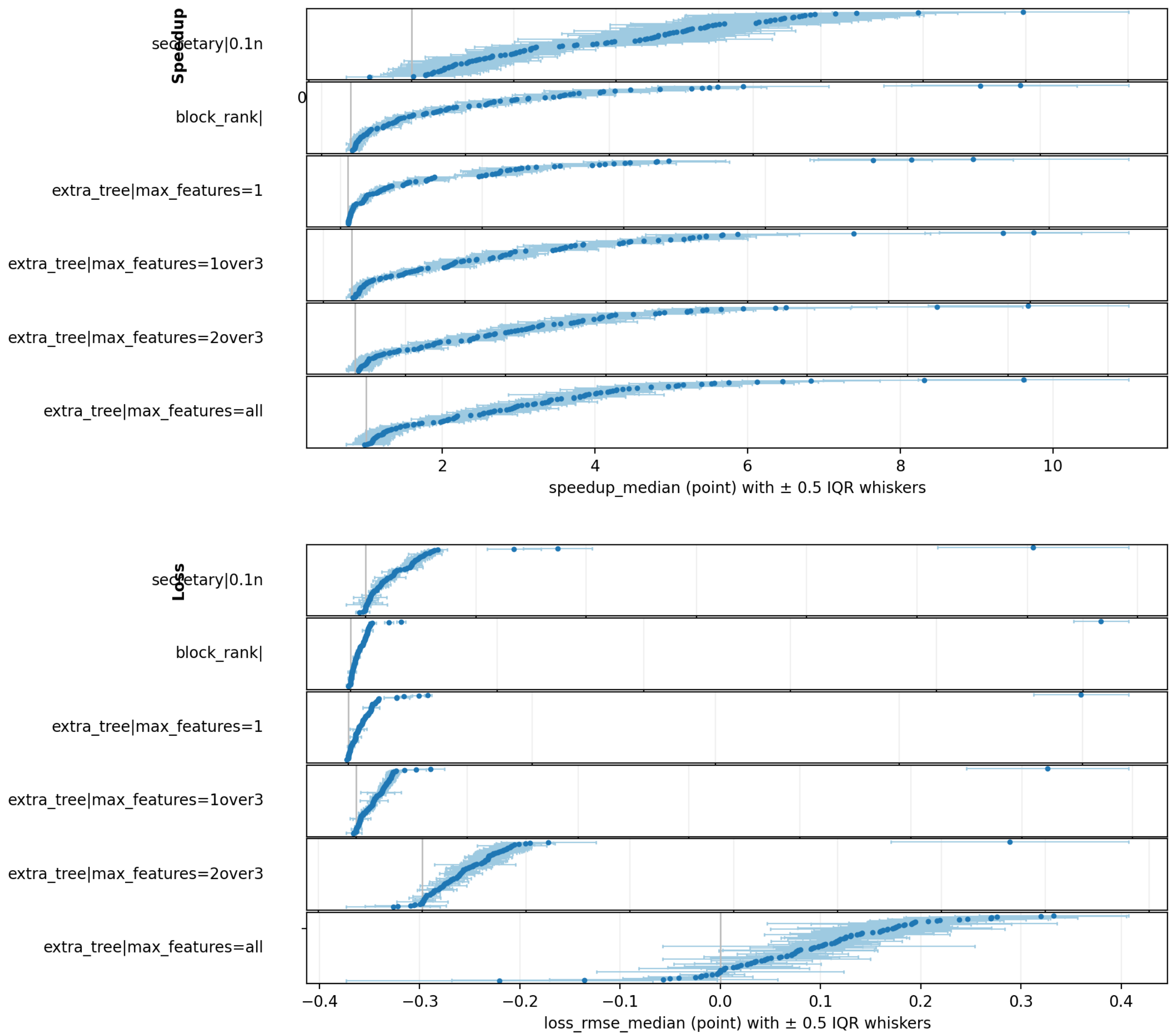

Figure S9: Within-dataset whisker summaries for regression. The upper half of the figure reports dataset-level median speedups for selected methods, and the lower half reports the corresponding dataset-level median RMSE losses. Each point represents one benchmark dataset, ordered within each method panel. Horizontal whiskers indicate ± 0.5 times the empirical interquartile range over the 50 repeated runs, thereby quantifying within-dataset stochastic variability. The figure makes explicit that methods with similar cross-dataset medians can nevertheless differ substantially in their run-to-run stability.

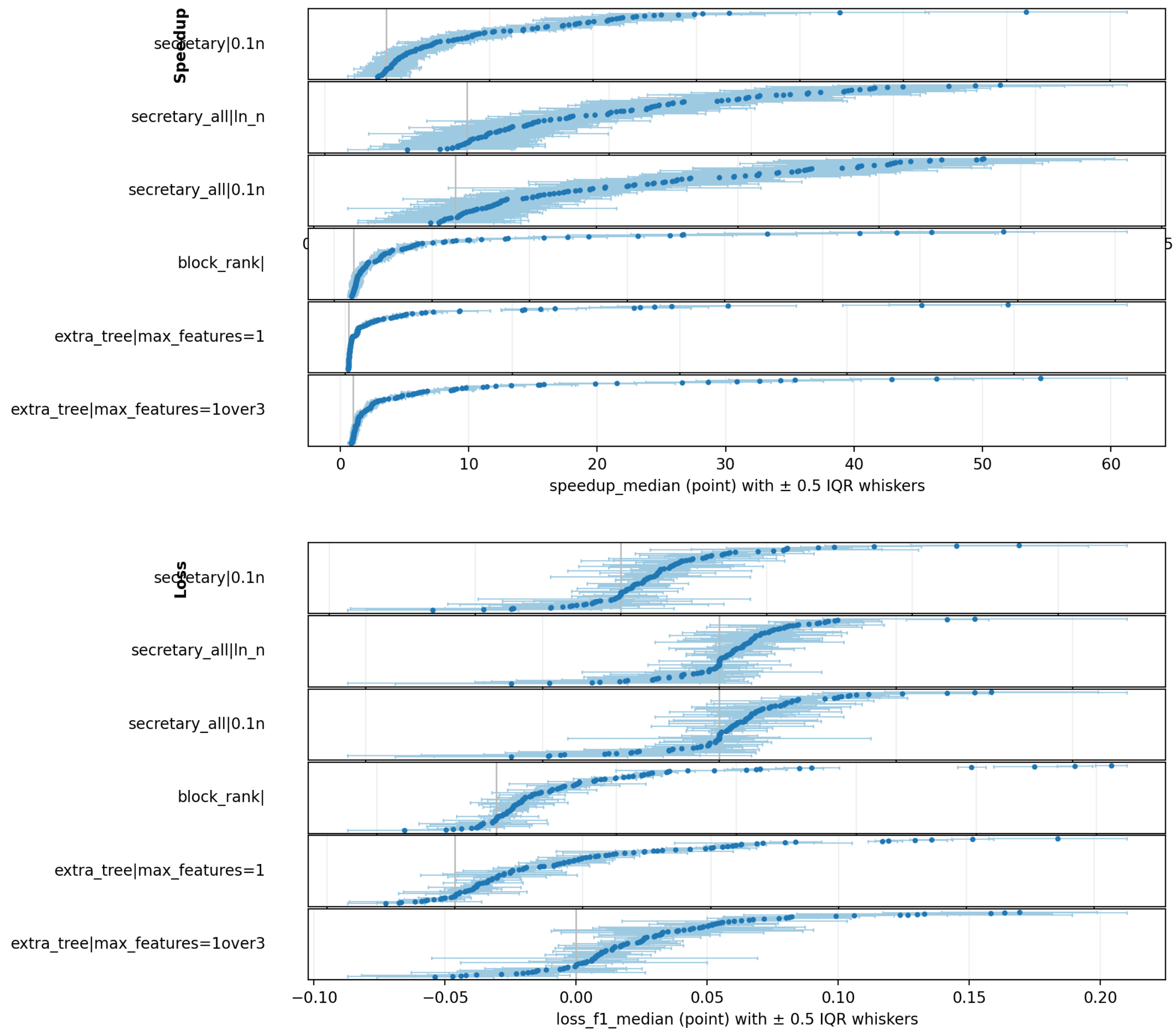

Figure S10: Within-dataset whisker summaries for classification with Gini impurity. As in Supplementary Figure S9, the upper half reports dataset-level median speedups and the lower half reports dataset-level median F1 losses for a selected subset of methods. Each point is one dataset and the whiskers indicate ± 0.5 times the empirical interquartile range across the 50 repeated runs. The figure shows that several methods achieve similar speedups but differ materially in the stability of their classification loss.

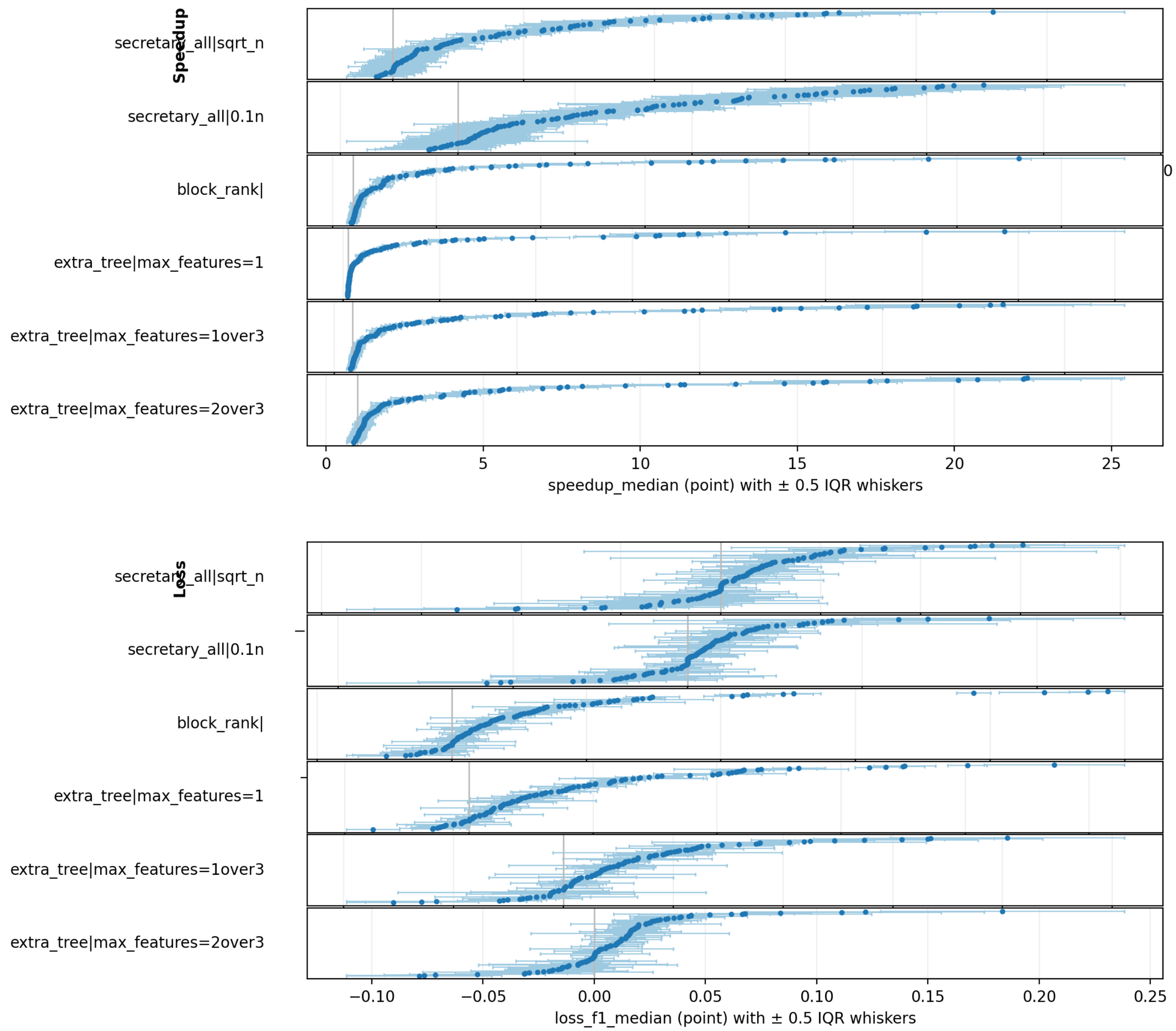

Figure S11: Within-dataset whisker summaries for classification with entropy impurity. The upper half reports dataset-level median speedups and the lower half reports dataset-level median F1 losses, with points corresponding to datasets and whiskers indicating ± 0.5 times the empirical interquartile range over the 50 repeated runs. Together with Supplementary Figure S10, this figure shows that the broad classification conclusions of the paper do not depend strongly on the choice between Gini and entropy impurity.

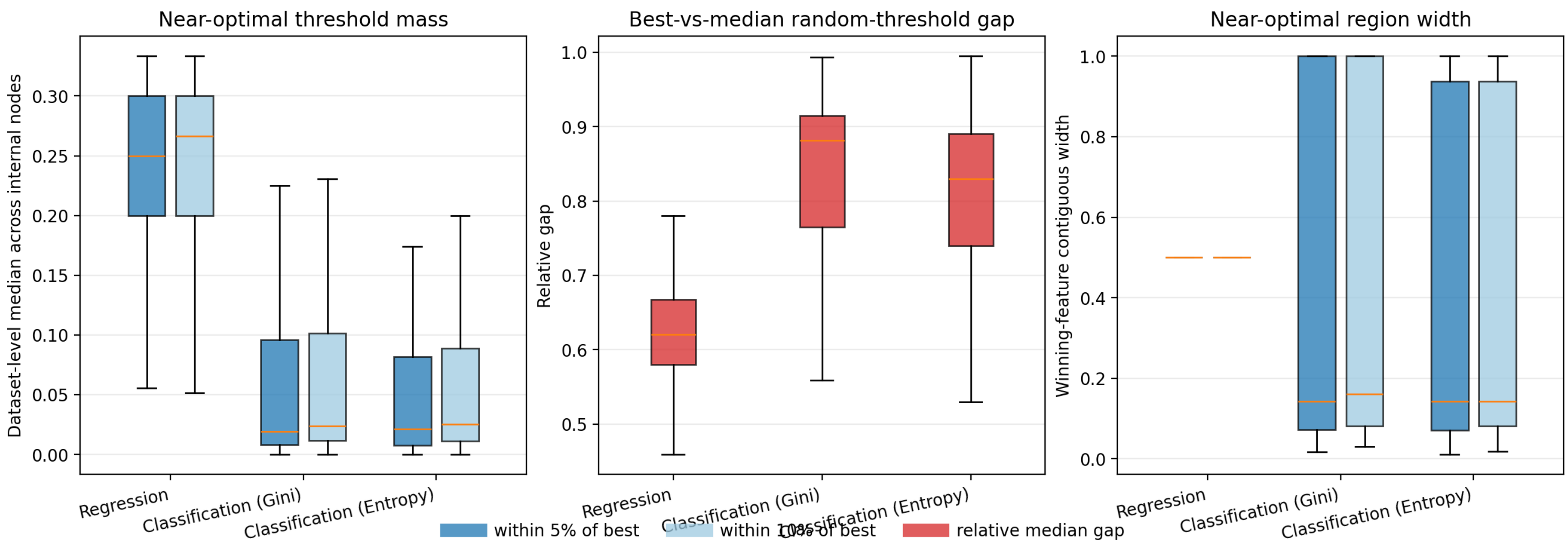

Figure S12: Exhaustive gain-landscape diagnostics by task. Each box summarizes one benchmark task using dataset-level medians of internal-node diagnostics computed under exhaustive split search. The left panel reports the proportion of admissible thresholds whose gain lies within 5% or 10% of the nodewise optimum, the middle panel reports the relative gap between the best gain and the median gain of a random admissible threshold, and the right panel reports the contiguous width of the near-optimal region on the winning covariate. The three diagnostics need not rank tasks in the same way; taken together, they show that the regression–classification asymmetry is not captured by a single scalar notion of gain-landscape flatness.

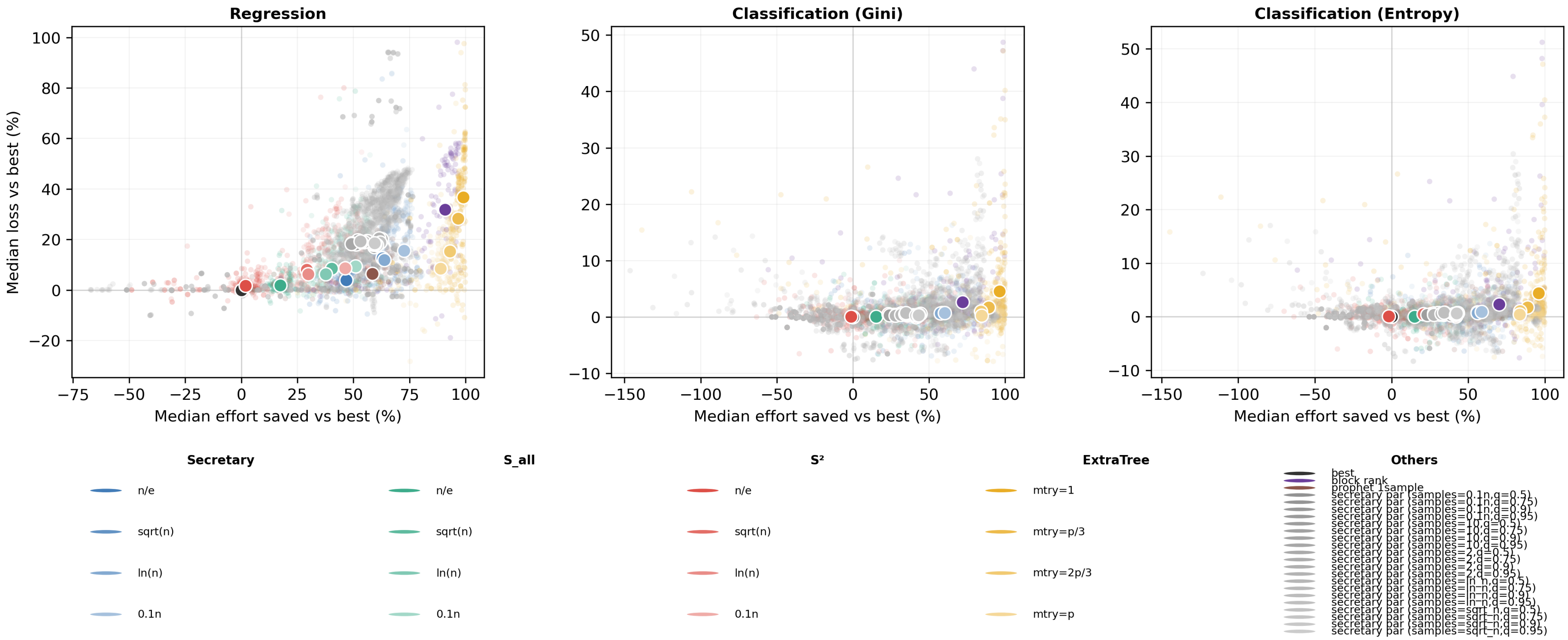

Figure S13: Effort–loss comparison by task. Each point represents one method variant summarized by the median over benchmark datasets. The horizontal axis reports the median percentage of gain evaluations saved relative to the exhaustive splitter, and the vertical axis reports the corresponding median predictive loss. Separate panels are shown for regression, classification with Gini impurity, and classification with entropy impurity. This figure complements the wall-clock Pareto plots by showing how closely time savings track the underlying reduction in exact split-gain computations.

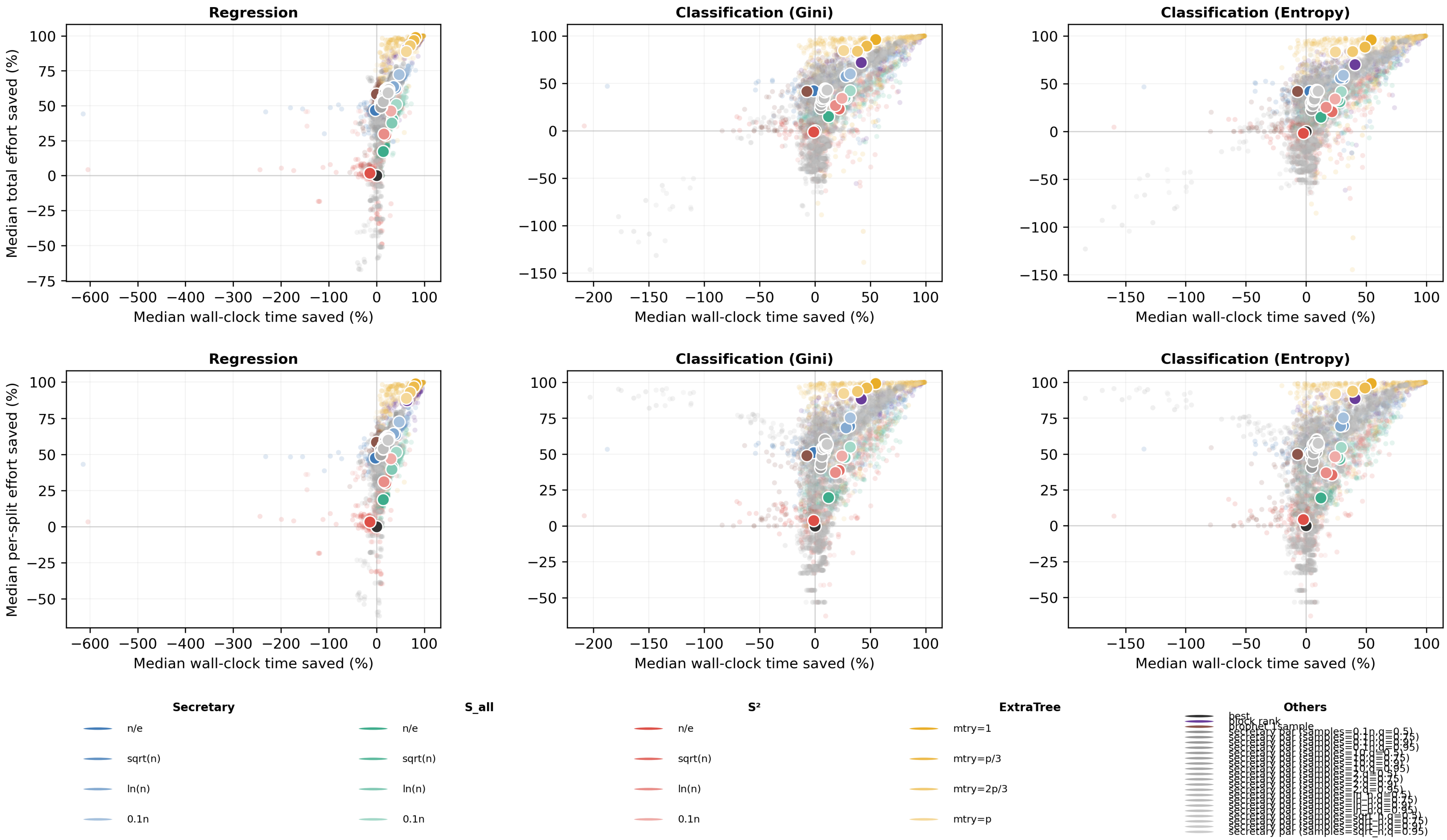

Figure S14: Calibration of wall-clock savings against split-search effort. The top row compares median percentage time saved with median percentage total effort saved, where total effort is measured through the number of gain evaluations per tree. The bottom row compares median percentage time saved with median percentage effort saved per split call. Separate panels are shown for regression, classification with Gini impurity, and classification with entropy impurity. These plots show which methods convert reduced split-search effort into actual runtime gains most efficiently, and which methods incur larger implementation overheads relative to their nominal effort budget.